GLP-1 Utilization & Health Needs Across U.S. States

IS 507 · Midterm Project Proposal · Group 1

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Github Repository ↗

Why this project?

"I am a data scientist at a public health organization tasked with examining whether GLP-1 prescription patterns align with healthcare needs across U.S. states. The available datasets include Medicaid GLP-1 prescription records, Medicaid enrollment data, and state-level obesity and diabetes prevalence data."

MOTIVATION

GLP-1 utilization may vary across states with similar health needs, raising questions about how Medicaid prescription patterns relate to obesity and diabetes prevalence.

RESEARCH QUESTION

Are U.S. states with Medicaid GLP-1 prescription rates below the levels predicted by their obesity and diabetes prevalence catching up during 2025?

POTENTIAL IMPACT

Flag states whose utilization patterns may warrant further public health investigation.

Related work: what earlier research found

PREVIOUS RESEARCH — DO ET AL. (2026)

Studied state-level GLP-1 use using 2023 prescription claims.

Found weak correlations between disease prevalence and Medicaid GLP-1 use.

HOW OUR PROJECT EXTENDS IT

Focus on 2025 rather than 2023.

Track changes in model-based utilization gaps from Q1 through Q4.

Examine which states move closer to predicted rates.

Takeaway: Health prevalence may matter, but it may not fully explain prescription patterns.

Prediction setup: inputs, target and error

X · INPUTS

- Obesity prevalence
- Diabetes prevalence
- Quarter (categorical)

Y · TARGET

GLP-1 prescriptions per Medicaid enrollee per quarter

$$Y = \frac{\text{Quarterly GLP-1 prescriptions}}{\text{Average quarterly Medicaid enrollment}}$$

Loss: Mean Squared Error (MSE)

MSE measures the average squared difference between predicted and actual GLP-1 prescription rates.

Three datasets, three distinct roles

01 Medicaid Drug Utilization — CMS

- GLP-1 prescription counts
- State × quarter
- Source: data.medicaid.gov

02 2025 Behavioral Risk Factor Surveillance System (BRFSS) — CDC

- Obesity & diabetes prevalence
- State × year
- Source: cdc.gov/brfss

03 Medicaid Enrollment — CMS

- Average Medicaid enrollment
- State × quarter
- Source: data.medicaid.gov

CDC Health Survey(BRFSS) → **Obesity & Diabetes Prevalence (X)**

Drug Prescriptions ÷ Medicaid Enrollment → **GLP-1 Prescription Rate (Y)**

What the raw data looks like

Medicaid Drug Utilization

State (str) | NDC (int) | Year (int) | Quarter (int) | Product Name (str)
Number of Prescriptions (float) | Suppression Used (bool)

5,308,547 rows × 15 columns · Numeric / categorical

	State	NDC	Year	Quarter	Suppression Used	Product Name	Number of Prescriptions
0	AK	00002143380	2025	4	False	TRULICITY	113.0
1	AK	00002143380	2025	4	True	TRULICITY	NaN
2	AK	00002143480	2025	4	False	TRULICITY	147.0

2025 Behavioral Risk Factor Surveillance System (BRFSS)

State Identifier (_STATE) | BMI Category (_BMI5CAT) | Diabetes Status (DIABETE 4) | Survey Weight (_LLCPWT) | Primary Health Insurance (PRIMINS2)

356,158 rows × 283 columns · Numeric / categorical

	_STATE	_BMI5CAT	DIABETE4	_LLCPWT	PRIMINS2
0	1.0	4.0	3.0	135.411848	3.0
1	1.0	2.0	1.0	190.376819	7.0
2	1.0	4.0	1.0	471.630521	3.0

Medicaid Enrollment

State Abbreviation (str) | State Name (str) | Reporting Period (int) Preliminary or Updated (str) | Final Report (str) | Total Medicaid Enrollment (float)

11,118 rows × 44 columns · Numeric / categorical

	State Abbreviation	State Name	Reporting Period	Preliminary or Updated	Final Report	Total Medicaid Enrollment
0	AK	Alaska	201309	U	Y	NaN
1	AK	Alaska	201706	P	N	179001.0
2	AK	Alaska	201706	U	Y	182080.0

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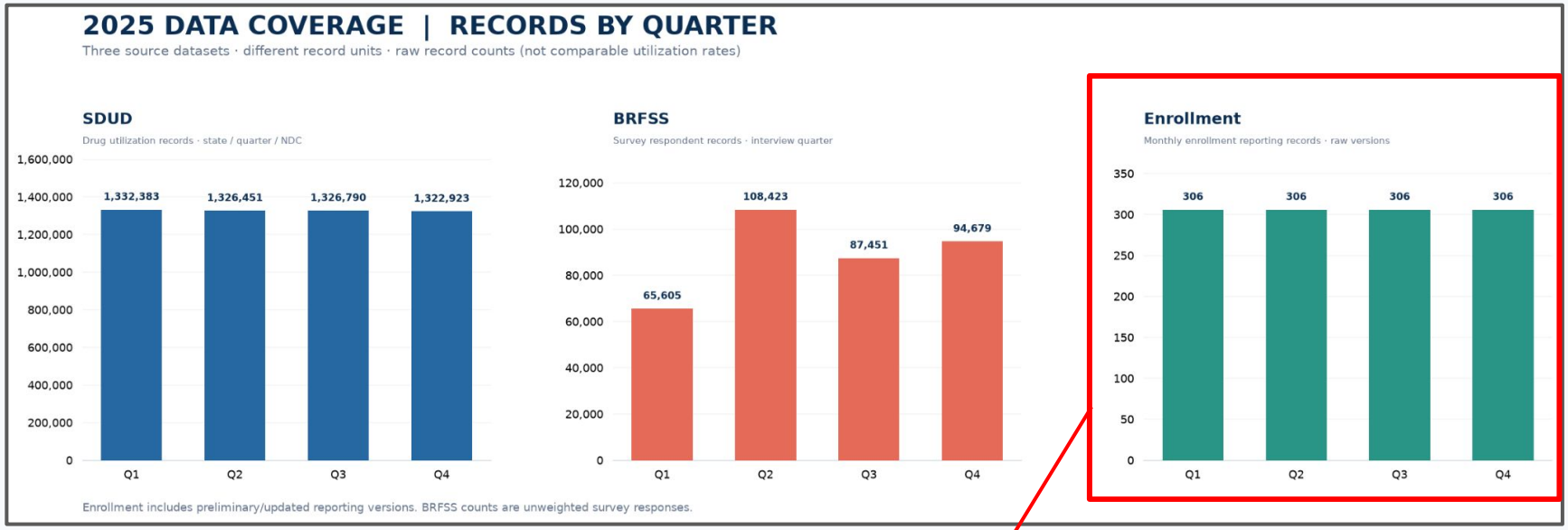
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What the raw data looks like



Why are Enrollment Records equal across quarters?

Each quarter has **306 records** (51 states/areas × 3 months × 2 reporting versions).

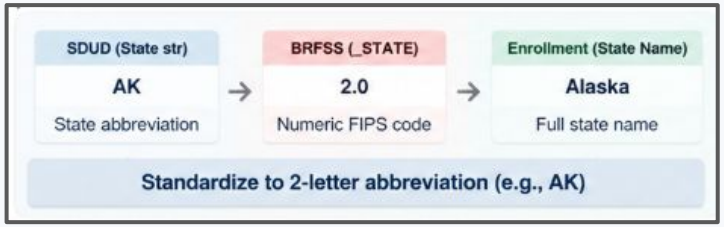
After retaining only final reports, **153 records per quarter (612 total)** remain, with no duplicates.

Note: These are reporting records, not enrollment counts.

Data Cleaning & Preparation

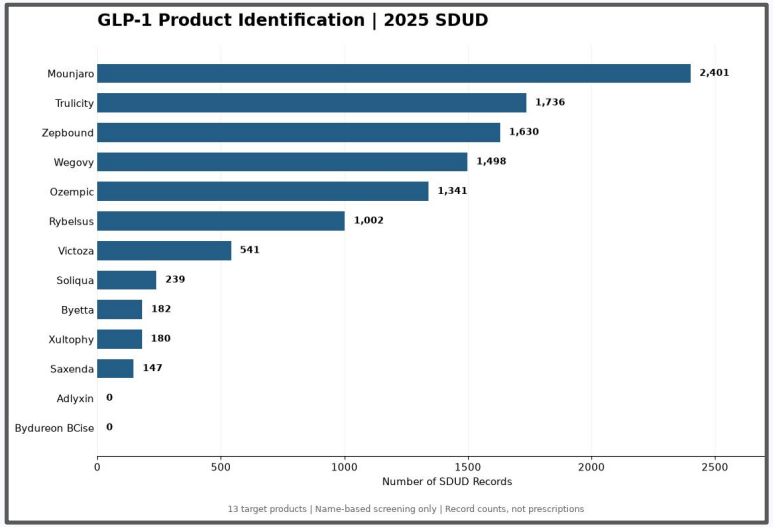
1 Select

Retain relevant variables; **standardize state codes and periods.**



2 Identify

Filter **GLP-1 products** using drug names and National Drug Code(NDC) information.



Data Cleaning & Preparation

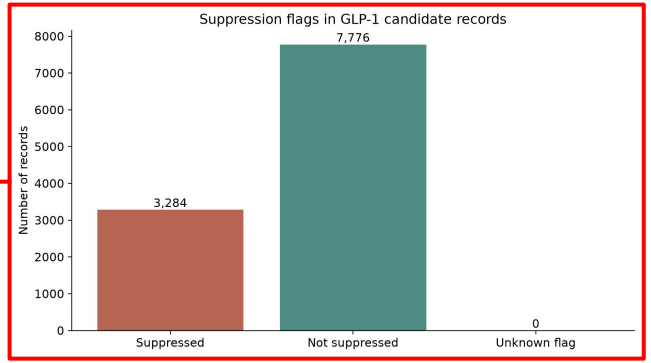
3 Clean

Review suppressed counts, missing values and BRFSS coded responses.

Missing Values in Key Variables (2025)

Selected variables with missing values in SDUD, BRFSS, and Medicaid Enrollment

Dataset	Variable	Missing Count	Missing (%)
SDUD (N = 5,306,991) 2025 records	Number of Prescriptions	**2,670,330**	**50.30%**
	Product Name	6,110	0.12%
BRFSS (N = 356,173) Respondents	_BMI5CAT	**32,562**	**9.14%**
	DIABETE4	6	0.00%
	PRIMINS2	2	0.00%
Enrollment (Final) (N = 5,508) 2025 final reports	Total Medicaid Enrollment	**51**	**0.92%**



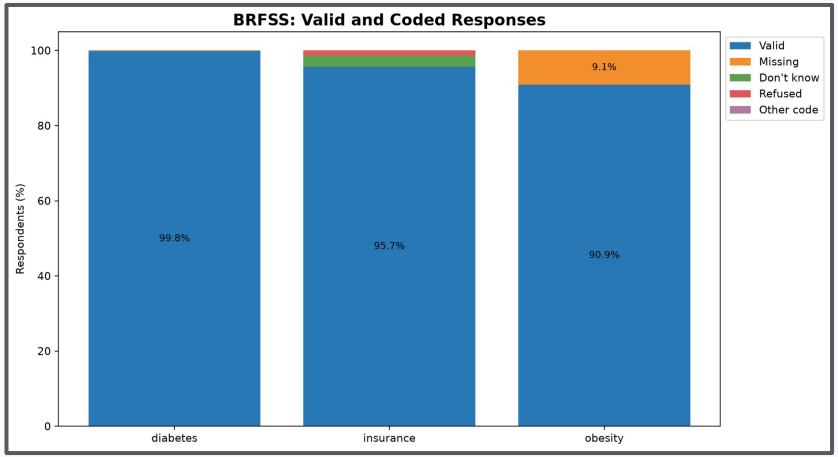
Key Finding: While **50.3% of all SDUD records** have suppressed prescription counts, **29.7% of GLP-1 candidate records** are suppressed.

These unavailable counts must not be treated as zero!

Data Cleaning & Preparation

3 Clean

Review suppressed counts, missing values and BRFSS coded responses.



4 Combine

Aggregate by state-quarter, apply survey weights and merge.

Final Analytical Dataset — Illustrative Structure

State	Quarter	Obesity (%)	Diabetes (%)	GLP-1 Rate
IL	Q1	X1	X2	Y1
IL	Q2	X1	X2	Y2
IL	Q3	X1	X2	Y3
IL	Q4	X1	X2	Y4

One row per state and quarter | X: Health indicators | Y: GLP-1 prescription rate

Potential challenges

Suppressed prescription counts

Inspect missingness; never replace suppression with zero.

29.7% of GLP-1 candidate records suppressed

Data quality

Handle missing and coded responses; use survey weights.

9.14% of BRFSS BMI category values missing

Population differences

Check Medicaid-specific BRFSS precision; disclose mismatch.

Time alignment

Align enrollment periods; annual health rates may stay fixed.

Interpretation boundary: utilization ≠ access, unique patients, or causal evidence.

Why multiple linear regression?

Primary Model:
MULTIPLE LINEAR REGRESSION

Interpretable: Easy to explain how each predictor relates to GLP-1 prescription rates.

Multiple predictors: Combines obesity prevalence, diabetes prevalence, and quarter.

Predictive value: Tests whether health indicators improve accuracy.

Potential Alternative Models

Ridge Regression: May improve stability when predictors are correlated.

Random Forest: May capture complex, nonlinear patterns.

Why start here?

Multiple Linear Regression provides an interpretable starting point for estimating expected GLP-1 prescription rates and identifying states with lower-than-expected utilization.

Baseline and fair model evaluation

BASELINE

Use the **training-state average prescription rate for each quarter**; do not use health indicators.

Why chosen? To test whether obesity and diabetes improve predictions beyond a simple quarterly average.

REGRESSION MODEL

Predict rates from obesity prevalence, diabetes prevalence and quarter.

State-Grouped Cross-Validation

Keep all four quarters of each state in one fold

Evaluation

Compare held-out MSE on identical folds. Use Out-of Fold predictions for gap analysis

Are lower-utilization states catching up?

Utilization gap = Actual prescription rate − Predicted prescription rate

NEGATIVE GAP

Actual GLP-1 prescription rates are lower than model-predicted rates.

CATCHING UP

The negative gap narrows toward zero between Q1 and Q4.

INTERPRETATION

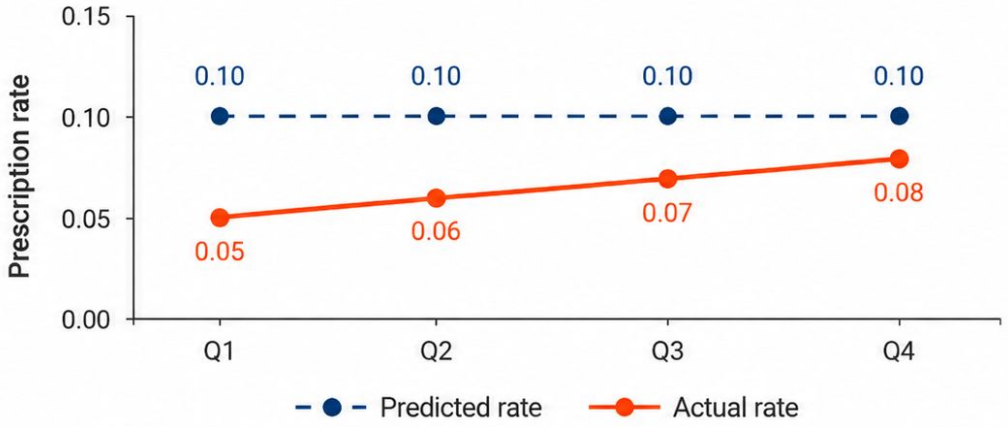
Compare actual and predicted rates over time to distinguish increasing utilization from changes in model expectations.

Our Goal: Identify states with initially negative utilization gaps and examine whether **those gaps narrow over 2025**.

Are lower-utilization states catching up?

Illustrative Example: A State Catching Up

Hypothetical GLP-1 prescription rates per Medicaid enrollee (illustrative values only)



Quarter	Predicted rate	Actual rate	Utilization gap
Q1	0.10	0.05	-0.05
Q2	0.10	0.06	-0.04
Q3	0.10	0.07	-0.03
Q4	0.10	0.08	-0.02

The negative gap narrows as actual utilization approaches the predicted level.

Expected Project Outcomes

A reproducible state-quarter analysis of Medicaid GLP-1 prescribing patterns

Identify low-utilization states · Assess whether gaps narrow · Highlight areas for further investigation

Our findings will highlight potential disparities, not establish their causes.

References & Sources

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